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A machine learning-based, decision support, mobile phone application for diagnosis of common dermatological diseases

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Abstract

Background The integration of machine learning algorithms in decision support tools for physicians is gaining popularity. These tools can tackle the disparities in healthcare access as the technology can be implemented on smartphones. We present the first, large-scale study on patients with skin of colour, in which the feasibility of a novel mobile health application (mHealth app) was investigated in actual clinical workflows.

Objective To develop a mHealth app to diagnose 40 common skin diseases and test it in clinical settings.

Methods A convolutional neural network-based algorithm was trained with clinical images of 40 skin diseases. A smartphone app was generated and validated on 5014 patients, attending rural and urban outpatient dermatology departments in India. The results of this mHealth app were compared against the dermatologists' diagnoses.

Results The machine–learning model, in an *in silico* validation study, demonstrated an overall top-1 accuracy of 76.93 ± 0.88% and mean area-under-curve of 0.95 ± 0.02 on a set of clinical images. In the clinical study, on patients with skin of colour, the app achieved an overall top-1 accuracy of 75.07% (95% CI = 73.75–76.36), top-3 accuracy of 89.62% (95% CI = 88.67–90.52) and mean area-under-curve of 0.90 ± 0.07.

Conclusion This study underscores the utility of artificial intelligence-driven smartphone applications as a point-of-care, clinical decision support tool for dermatological diagnosis for a wide spectrum of skin diseases in patients of the skin of colour.

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Conflict of interests

Jyoti Mathur and Sharad Kumar are direct beneficiaries of any profits acquired by the smartphone application developed at Nurithm Labs Private Limited and Vikas Chouhan is a salaried employee at Nurithm Labs Private Limited.

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Introduction

Skin diseases are the fourth leading cause of non-fatal disease burden across 188 countries in both low and high-income countries.[1,2] There is less than one dermatologist per 100 000 people in India and less than three dermatologists per 100 000 of the population in the United States and the ratios are expected to decline further.[3,4] In a survey by the American Academy of Dermatology, 33% of dermatologists (48% in rural areas) reported that the number of dermatologists in their community is insufficient to meet patient demand.[5] Furthermore, appointment wait times for dermatologists can range from 4–12 weeks in the US accounting for 42.7% of all primary care physician (PCPs) consultations being dermatology-related.[6] However, across the globe, the accuracy of PCPs has been reported to be 40–60% for skin disease diagnoses.[7–9] In a study by Tran et al, the referring PCPs' diagnosis agreed with the dermatologists' clinical diagnosis and histology in only 42% of cases.[10] Furthermore, unnecessary PCP referrals to dermatologists increase the burden on the tertiary healthcare systems. The delays in getting appointments, misdiagnoses and the high cost of getting correct treatment can lead to adverse repercussions.[11]

The recent advances in mobile phone technology, its high penetration, along with the development of healthcare-focused artificial intelligence (AI) algorithms provide a fertile ground to enable a synergistic, low-cost solution to alleviate the healthcare inequity that exists all over the world.[12] According to a recent report, there are approximately 3.2 billion smartphone users around the world.[13] An AI-based mobile phone application (app) that provides diagnostic decision support to PCPs can empower them to diagnose diseases more accurately and hence immensely increase access to affordable care in underserved areas.[14] However, so far, smartphone applications using AI-based analysis have not consistently demonstrated promise in terms of accuracy and are associated with a high likelihood of false negatives. For instance, in a Cochrane review, data for four such mobile phone applications (available as of 2016) that use AI algorithms for melanoma classification were analysed and the sensitivities of the apps ranged from 7% to 73% and specificities from 37% to 94%.[15]

There are no previous reports of validation studies of AI-based algorithms related to dermatology in clinical settings.

Material and methods

The study aimed to develop a convolutional neural network (CNN)-based algorithm trained with clinical images of 40 different skin diseases. A user-friendly, smartphone app was also generated, and a clinical validation study on 5014 patients was done by physicians in urban, rural primary care and tertiary care settings. The app's analysis of a single image of the lesion was compared to the consensus diagnosis made by two board-certified dermatologists. Only patients, for whom the treating board-certified dermatologist was confident of the diagnosis and the diagnosis was cross-verified by another board-certified dermatologist, were included. If there was no consensus then the patient was excluded from the study.

Generation of the CNN-based algorithm

The CNN architecture used was a modified version of Densenet-161 with optimized augmentation and inference pipelines.[16] These optimizations were derived through rigorous experiments and were attuned to clinical skin images. All training and testing images were annotated and segmented so a bounding box surrounded lesion of interest and irrelevant features such as rulers and surgical markings were discarded. The 17 718 raw images were sourced from public databases (http://www.hellenicdermatlas.com/en and http://www.danderm.dk/atlas), after obtaining permission from them, as well as images from dermatologists in India. Of these, 310 images were discarded during the preprocessing stage due to poor resolution or multiple lesions. Out of the remaining 17 408 images, 1990 images belonged to the non-specific category and 15 418 images were within the 40 selected disease categories (Table 1). These images were split into five equal parts (folds) and five iterations of training and validation were performed in a manner so that within each iteration, a different fold of the data was held-out for validation while the remaining fourfolds were used for learning. The training images were 12 350 and testing images were 3068. Since the dataset was imbalanced, a custom modified version of the weighted focal loss function was used to train the network. The weights for each class were calculated using the inverse sample size method. Our test images were either in-distribution images i.e. images within 40 training classes or out-of-distribution sets that we named as non-specific set.

We initially used Inception-v4, which was one of the most popular CNN models at the time. For 10 diseases, a sensitivity of around 85% was achieved, but for the 20-disease class model, this fell to around 65%. We experimented with many parameters like learning rate and choice of optimizer etc. with no significant improvements. It was felt that as the architectures became deeper, there could be problems related to over-fitting. According to Huang *et al.*, convolutional networks can be substantially deeper, more accurate and efficient to train if they contain shorter connections between layers close to the input and those close to the output.[17] Their implementation of this architecture is called DenseNet. In this architecture, to ensure maximum information flow between layers, each layer obtains additional inputs from all preceding layers and passes on its own feature-maps to all subsequent layers.[17] We found that the DenseNet architecture provided better results than the other contemporary architectures such as Inception-Resnet-V2, Resnets-50, Resnet-101 and Resnet-152. We tried several DenseNet variants such as Densenet-169, Densenet-121, Densenet-161 and densenet-201, but selected Densenet-161 as it achieved the best balance of sensitivity versus specificity for the 40 disease classes. In future, as the disease list expands and as newer CNN architectures evolve, this search for an optimal model will need to be revisited.

Our optimizations included three stages. In the preprocessing stage, via iterative experiments, we identified a set of specific image processing and augmentation algorithms that improve the overall model performance. We determined that for skin lesions random image crop selection within a targeted area range, slight colour balance adjustment, image flips (vertical and horizontal) and 90° rotations lead to an improvement in results. Few algorithms like heavy colour balance adjustment, random rotations lead to no improvements or even slight degradation in performance. We also did several experiments with normalization algorithms and selected the best performing Gaussian normalization algorithm on our dataset. In the postprocessing stage for prediction, we used several copies of the same model to generate the final prediction. The raw outputs from each of the model

Table 1 shows number of images and diagnostic parameters from fivefold algorithmic validation of 41 skin conditions

S.S	Disease class	*N*, Private	*N*, Public	AUC-model	Sensitivity	Specificity	PPV	NPV
1	Acne	337	240	0.98	86.23 ± 3.26	99.56 ± 0.13	86.32 ± 4.22	99.46 ± 0.19
2	Actinic keratosis	249	372	0.97	76.49 ± 3.25	99.04 ± 0.22	82.83 ± 3.72	98.85 ± 0.19
3	Alopecia	109	152	0.97	78.38 ± 6.25	99.74 ± 0.09	86.60 ± 5.39	99.65 ± 0.08
4	Anogenital warts	86	96	0.95	80.22 ± 1.96	99.92 ± 0.04	85.69 ± 6.51	99.73 ± 0.10
5	Basal cell carcinoma	365	506	0.97	77.77 ± 3.58	99.10 ± 0.13	83.89 ± 1.09	98.50 ± 0.27
6	Bowen's disease	130	179	0.92	61.31 ± 10.78	99.60 ± 0.14	79.31 ± 8.58	99.06 ± 0.11
7	Bullous pemphigoid	54	92	0.91	58.22 ± 13.60	99.90 ± 0.05	82.86 ± 8.45	99.60 ± 0.14
8	Candidiasis	118	187	0.94	65.24 ± 6.39	99.54 ± 0.13	77.47 ± 4.65	99.35 ± 0.11
9	Chicken pox	77	108	0.94	72.43 ± 6.99	99.77 ± 0.07	80.94 ± 9.67	99.59 ± 0.12
10	Discoid lupus erythematosus	85	134	0.89	49.78 ± 6.70	99.83 ± 0.07	73.06 ± 6.84	99.36 ± 0.07
11	Eczema	560	336	0.92	57.52 ± 3.37	99.00 ± 0.12	66.20 ± 3.98	98.30 ± 0.15
12	Fixed drug eruption	65	99	0.94	73.26 ± 7.43	99.90 ± 0.07	91.03 ± 7.59	99.80 ± 0.11
13	Herpes zoster	98	153	0.96	79.28 ± 6.70	99.87 ± 0.10	90.83 ± 0.68	99.70 ± 0.08
14	Hidradenitis suppurativa	96	82	0.96	79.49 ± 11.56	99.93 ± 0.02	88.23 ± 4.82	99.73 ± 0.10
15	Ichthyosis	80	94	0.94	76.90 ± 3.57	99.97 ± 0.03	88.53 ± 4.65	99.64 ± 0.06
16	Impetigo and Pyodermas	198	221	0.95	78.76 ± 3.81	99.76 ± 0.12	84.55 ± 4.86	99.19 ± 0.11
17	Keloids/Hypertrophic scar	88	126	0.96	74.55 ± 7.06	99.96 ± 0.04	91.15 ± 2.86	99.64 ± 0.11
18	Keratoacanthoma	132	173	0.98	79.85 ± 4.46	99.76 ± 0.08	87.67 ± 5.75	99.44 ± 0.07
19	Lichen planus	205	210	0.94	71.32 ± 2.15	99.65 ± 0.06	78.96 ± 8.07	99.31 ± 0.09
20	Lichen sclerosus	83	137	0.93	73.18 ± 8.41	99.88 ± 0.08	87.50 ± 7.48	99.68 ± 0.10
21	Melanocytic nevi/Mole	153	134	0.95	77.02 ± 8.37	99.73 ± 0.11	84.78 ± 6.35	99.65 ± 0.12
22	Melanoma	99	211	0.95	77.74 ± 2.10	99.81 ± 0.08	82.41 ± 3.91	99.45 ± 0.09
23	Melasma	88	77	0.99	87.50 ± 6.25	99.78 ± 0.08	88.00 ± 6.43	99.94 ± 0.06
24	Milia	52	60	0.94	77.86 ± 12.92	99.95 ± 0.02	85.43 ± 10.1	99.75 ± 0.10
25	Molluscum contagiosum	48	234	0.97	79.75 ± 7.72	99.81 ± 0.08	85.41 ± 4.63	99.58 ± 0.13
26	Pemphigus	99	137	0.94	66.51 ± 1.96	99.76 ± 0.07	80.81 ± 4.33	99.59 ± 0.07
27	Pityriasis rosea	95	155	0.95	76.80 ± 5.21	99.75 ± 0.08	82.42 ± 4.68	99.61 ± 0.06
28	Pityriasis versicolour	67	108	0.95	69.71 ± 5.92	99.83 ± 0.07	83.51 ± 5.90	99.64 ± 0.07
29	Psoriasis	452	421	0.94	68.00 ± 5.06	99.18 ± 0.13	80.40 ± 6.34	98.41 ± 0.20
30	Rosacea	184	255	0.98	90.17 ± 4.40	99.62 ± 0.13	92.44 ± 2.47	99.70-±0.06
31	Seborrhoeic keratosis	211	238	0.97	74.39 ± 3.92	99.39 ± 0.17	86.66 ± 3.35	99.47 ± 0.16
32	Squamous cell carcinoma	157	212	0.92	58.29 ± 4.65	99.41 ± 0.08	75.74 ± 5.82	99.03 ± 0.06
33	Tinea capitis	4	153	0.96	79.00 ± 4.84	99.91 ± 0.06	80.20 ± 7.45	99.68 ± 0.04
34	Tinea cruris, corporis or faciei	603	493	0.96	75.41 ± 2.35	98.90 ± 0.10	86.00 ± 3.89	98.20 ± 0.15
35	Tinea manuum	1	130	0.98	71.77 ± 3.28	99.75 ± 0.07	74.63 ± 3.09	99.73 ± 0.06
36	Tinea pedis	0	258	0.98	82.94 ± 4.02	99.62 ± 0.12	87.26 ± 5.53	99.64 ± 0.07
37	Tinea unguium	1399	45	0.99	96.10 ± 1.33	99.80 ± 0.09	96.74 ± 1.79	99.41 ± 0.12
38	Urticaria	77	90	0.92	70.10 ± 7.58	99.91 ± 0.04	81.77 ± 6.39	99.73 ± 0.08
39	Viral warts	222	206	0.91	59.83 ± 7.31	99.55 ± 0.13	83.50 ± 6.83	99.07 ± 0.16
40	Vitiligo/Leucoderma	195	145	0.98	84.10 ± 6.84	99.79 ± 0.11	89.88 ± 1.65	99.69 ± 0.12
41	Normal skin	80	768	0.99	98.33 ± 0.74	99.90 ± 0.05	97.85 ± 0.72	99.80 ± 0.12
	Overall Top-1 accuracy			**0.95 ± 0.02**	**76.93 ± 0.88**			

copies were then combined to generate final predictions. We additionally made custom changes in the loss function. Focal loss is used for multi-class classification whereas log loss for binary classification. We implemented a new loss function that led to better results from either of them individually. Our custom loss function is a hybrid implementation of focal loss and log loss.[18,19] We also accounted for class imbalances. The mathematical expression of the loss function used by us is described below.

$$\mathrm{FL}(p_t) = -\alpha_t (1-p_t)^{\gamma} \log(p_t)$$

$$\mathrm{LL}(p) = -(y\log(p) + (1-y)\log(1-p))$$

$$\mathrm{FCL}(p_t) = -\alpha_t (y_t(1-p_t)^{\gamma} \log(p_t) + (1-y_t)p_t^{\gamma} \log(1-p_t))$$

FL = Focal loss; LL = Log Loss; FCL = Our loss function implementation.

We used an input resolution of 224 × 224 for the model. The image is resized to this resolution after the automatic crop selection algorithm completes. Due to this algorithm, we got lots of random crops per input image effectively leading to dataset enhancement. After resizing (using cubic interpolation), the rest of the augmentation algorithms were applied and at the end, each of the generated crop was normalized using our preselected normalization algorithm before feeding into the model.

For Densenet-161, we used a batch size of 24–32 (depending on the GPU), also adjusted the learning rate (between 0.0001 and 0.001) accordingly. We used the RMSProp optimizer and an exponential decaying learning rate scheduling algorithm for the training process.

Generation of app based on an algorithm

iOS and Android mobile phone apps were generated. In this app, the first three disease predictions with their respective probabilities were shown, which were further normalized across themselves as $p_1/(p_1 + p_2 + p_3)$, $p_2/(p_1 + p_2 + p_3)$ and $p_3/(p_1 + p_2 + p_3)$. This app uses single image input either taken from the mobile phone camera or already saved on the phone.

Determination of model and app performance

Model performance was measured in terms of per disease sensitivity, specificity, positive predictive values (PPV) and negative predictive values (NPV), area-under-the-curve (AUC) and mean AUC. Mean values with standard deviation were shown for disease-specific sensitivities, specificities, PPV, NPV and overall accuracy of model and app with 95% confidence intervals (CI) were calculated. Receiver operating characteristics (ROC) curves were plotted using probability scores for each of the disease classes by varying the cutoff threshold as described by Pedregosa *et al.*[20] The threshold varied from 0 to 1 in equally spaced intervals and the sensitivity and specificity for each disease were measured as they changed. We used this data to calculate the true positive rate (same as sensitivity) and the false-positive rate (1 − specificity). ROC curve was the plot of the true positive rate versus the false-positive rate.

Clinical validation study

We tested our mobile app against dermatologists' diagnosis in three different clinical settings, namely, at a tertiary care centre

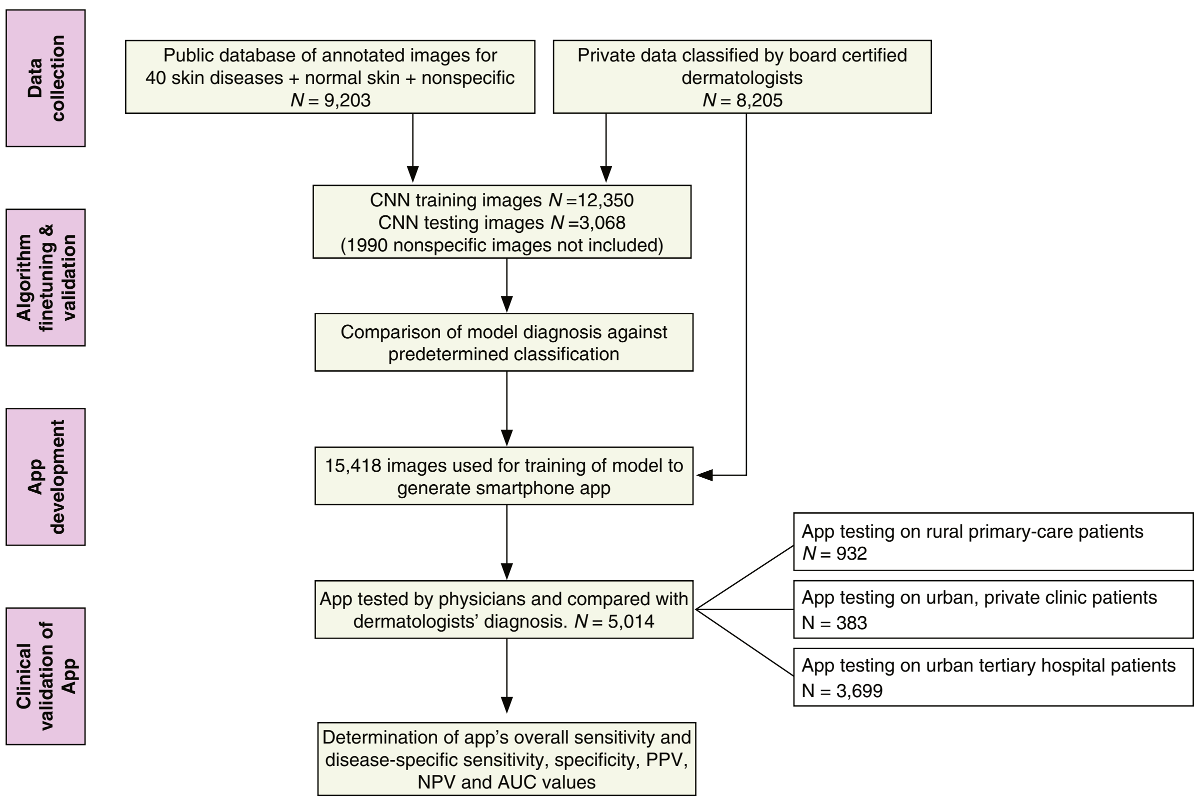


Figure 1 This flowchart depicts the different steps in data collection, algorithm generation, algorithm testing, app generation and app validation in clinical studies.

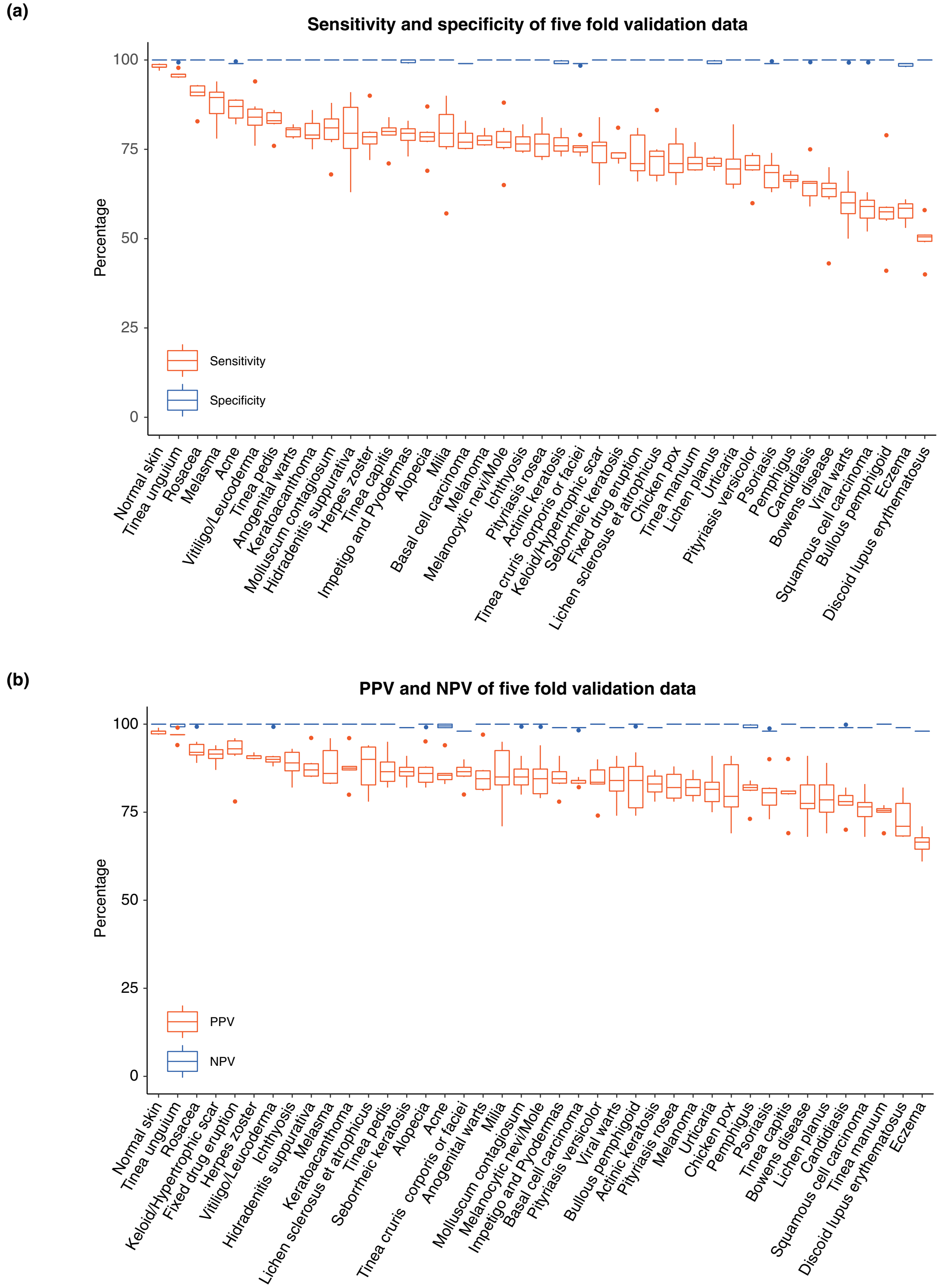


Figure 2 (a) Boxplot of disease-specific sensitivities and specificities. (b): Boxplot of disease-specific PPV (positive predictive value) and NPV (negative predictive value) for algorithmic validation of 41 skin conditions. The whiskers represent maximum and minimum values; the box edges represent the third and first quartiles; the horizontal lines within the box depict median values and the dots denote outliers.

Table 2 shows th number of patients, AUC, Top-1/Top-3 sensitivity, specificity, positive predictive and negative predictive values for combined clinical data from three different clinical settings.

S.no.	Disease class	Number of patients	AUC-clinic	Top-1 sensitivity % (CI)	Top-3 sensitivity %	Top-1 specificity %	Top-1 PPV	Top-1 NPV
1	Acne	592	0.98	89.86 (87.15–92.18)	97.97	98.19	86.92	98.64
2	Alopecia	184	0.97	88.11 (82.55–92.36)	96.22	99.28	82.32	99.54
3	Anogenital warts	34	0.85	57.14 (39.35–73.68)	71.43	99.70	57.14	99.70
4	Basal cell carcinoma	123	0.98	93.55 (87.68–97.17)	97.58	98.51	61.38	99.83
5	Bowen's disease	11	0.77	45.45 (16.75–76.62)	54.54	99.90	50.00	99.88
6	Bullous pemphigoid	16	0.73	35.29 (14.21–61.67)	47.06	100.00	100.00	99.78
7	Candidiasis	16	0.81	41.18 (18.44–67.10)	64.71	99.80	41.18	99.80
8	Discoid lupus erythematosus	47	0.90	60.42 (45.27–74.23)	83.33	99.20	42.03	99.61
9	Eczema	432	0.87	54.29 (49.46–59.10)	81.90	97.58	67.83	95.78
10	Fixed drug eruption	12	0.87	41.67 (15.16–72.33)	75.00	99.70	25.00	99.86
11	Herpes zoster	38	0.89	66.67 (49.78–80.91)	79.49	99.60	56.52	99.74
12	Hidradenitis suppurativa	32	0.97	93.94(79.78–99.26)	93.94	99.62	62.00	99.96
13	Ichthyosis	21	0.93	81.81 (59.72–94.81)	86.36	99.78	62.07	99.92
14	Impetigo and Pyodermas	109	0.89	57.27 (47.48–66.66)	82.72	99.04	57.27	99.04
15	Keloids/Hypertrophic scar	128	0.91	73.64 (65.16–81.00)	84.50	99.44	77.87	99.30
16	Lichen planus	170	0.88	69.59 (62.10–76.38)	84.80	95.66	36.17	98.89
17	Lichen sclerosus et atrophicus	17	0.86	44.44 (21.53–69.24)	72.22	99.90	61.54	99.80
18	Melanocytic nevi/Mole	138	0.83	43.17 (34.80–51.83)	67.63	99.79	85.71	98.40
19	Melasma	277	0.96	83.80 (80.00–87.89)	94.36	98.48	76.77	99.02
20	Milia	36	0.94	59.46 (43.00–75.25)	89.20	99.82	70.97	99.70
21	Molluscum contagiosum	51	0.98	84.62 (71.92–93.11)	96.15	98.93	45.36	99.84
22	Pemphigus	102	0.92	67.96 (58.04–76.81)	85.44	99.00	58.82	99.32
23	Pityriasis rosea	20	0.90	52.38 (29.78–74.29)	80.95	99.84	57.90	99.80
24	Pityriasis versicolour	77	0.95	75.64 (64.60–84.65)	92.31	99.13	57.84	99.61
25	Psoriasis	394	0.93	77.67 (73.22–81.68)	93.15	96.08	62.84	98.05
26	Rosacea	11	0.91	41.67 (15.16–72.33)	83.33	99.82	35.71	99.86
27	Seborrhoeic keratosis	41	0.85	52.38 (36.42–68.00)	71.43	99.57	50.00	99.60
28	Squamous cell carcinoma	13	0.96	78.57 (49.20–95.34)	92.86	99.64	37.93	99.94
29	Tinea capitis	32	0.87	54.55 (36.35–71.89)	75.76	99.90	78.26	99.70
30	Tinea cruris, corporis or faciei	545	0.95	81.69 (78.18–84.84)	96.22	96.75	75.47	97.73
31	Tinea pedis	10	0.75	40.00 (12.16–73.76)	50.00	99.90	44.44	99.88
32	Tinea unguium	64	0.96	76.92 (64.1–86.47)	92.31	99.70	76.92	99.70
33	Urticaria	60	0.91	55.74 42.46–68.45)	81.97	99.78	75.56	99.46
34	Viral warts	147	0.93	70.94 (62.92–78.11)	89.87	98.11	53.30	99.11
35	Vitiligo/ Leucoderma	254	0.98	92.94 (89.07–95.76)	97.65	98.76	80.07	99.62
36	**Overall/Mean ± S.D./ 95% C.I.**	**4254**	**0.90 ± 0.07**	**75.07 (73.75–76.36)**	**89.62 (88.67–90.52)**	**99.11 (99.06–99.15)**	**61.46 (55.77–67.14)**	**99.35 (99.07–99.62)**

(All India Institute of Medical Sciences, New Delhi), an urban private practice (Gurugram, Haryana, India) and a rural health centre (Jhajjar, Haryana, India). The three different centres were chosen to represent the diversity in the patient populations that consult physicians in urban versus rural areas. The study was approved by the Ethics Committee of All India Institute of Medical Sciences. We included consecutive patients presenting to a particular investigator in the Outpatient Department, where the dermatologists were confident of the diagnosis, either by clinical examination, laboratory investigation, and/or histopathological examination. Dermatologists were blinded to the app results. A single representative image from each patient was tested on the app by a separate physician.

Results

The overall scheme for this study is outlined in Figure 1.

In silico model validation

The performance of the CNN model for the diagnosis of 40 skin diseases showed a top-1 overall accuracy of 76.93 ± 0.88% and AUC of 0.95 ± 0.02 (Table 1). High specificities and NPV were observed for all disease classes (Table 1, Fig. 2a, and b). The sensitivities and PPV were more varied but 34/40 skin diseases showed high PPVs in 80–99% range. AUC values and ROC plots for different diseases in model validation and clinical study are given in Supplementary Figure S1.

Clinical validation

The app was tested on 5014 patients (3699 from tertiary care, 383 from urban private practice, and 932 from rural primary care). Out of the 5014 patients, 760 patients' images were in the nonspecific (out-of-distribution) category. The mean AUC was 0.90 ± 0.07, top-1 overall accuracy 75.07% (CI = 73.75–76.36) and top-3 accuracy 89.62% (CI = 88.67–90.52) for combined data from all centres (Table 2, Fig. 3a). The mean top-1 specificity was 99.11% (CI = 99.06–99.15). Small differences were noted in the overall accuracies of urban, rural primary and tertiary hospital patients, with the urban practice showing the highest and tertiary care centres showing the lowest accuracy (Fig. 3a). We do not present data for actinic keratosis, chickenpox, keratoacanthoma, melanoma and tinea manuum because of less than 10 patients for each of these disease classes. The most significant sensitivities were observed for diagnosis of hidradenitis suppurativa, BCC, vitiligo, acne, alopecia, molluscum contagiosum, melasma, ichthyosis and tinea corporis, cruris or faciei. Bowen's disease, bullous pemphigoid, candidiasis, fixed drug eruption, eczema, lichen sclerosus, melanocytic naevus, rosacea and tinea pedis had lower accuracy across all sites (Figs 3b and 4a and b). The sensitivity of diseases with larger sample sizes was generally greater except for eczema and melanocytic nevi (Fig. 3b).

The top-3 sensitivity of most diseases, including eczema, was high (Figs 3b and 4b). Analyses of the confusion matrix showed that many eczema lesions were misdiagnosed as psoriasis followed by lichen planus and tinea corporis, cruris or faciei (Fig. 3 a). Most of the skin cancers were detected in top-3 diagnoses as shown in the confusion matrix (Fig. 4b).

Discussion

This cross-sectional study is the first, large-scale, clinical validation study of an AI-driven app involving a wide spectrum of common dermatological diseases on the skin of colour. Moreover, this work shows the potential of an app to be successfully incorporated into clinical workflows by physicians using ordinary smartphones.

Nearly all previously published work on automated skin disease detection has focused on *in silico* algorithmic validation.[21] There is also little published literature demonstrating high-efficiency neural networks for multi-disease classification. AI-based algorithms, particularly CNN, have been successfully applied in skin lesion classification, mainly for skin cancers. However, only a few researchers have used clinical/macroscopic images for multi-class skin disease identification, representative of a real-world clinical scenario. In a study by Esteva *et al*., CNN performed *at par* with dermatologists for classification of 9 skin disease categories, skin cancers and their precursors, using clinical images with an accuracy of 55.4%.[22] Mendes *et al*. generated a CNN classification model for 12 skin cancers and related benign and premalignant skin conditions and achieved a total accuracy of 78%.[23] Han *et al*. reported a mean AUC of 0.91 and top-1 accuracy between ~55–57% on the same set of 12 diseases and, interestingly, found a substantial difference in specificities of skin cancer between an Asian skin dataset and Caucasian skin dataset.[24] More recently, Liu *et al*. developed a deep learning system

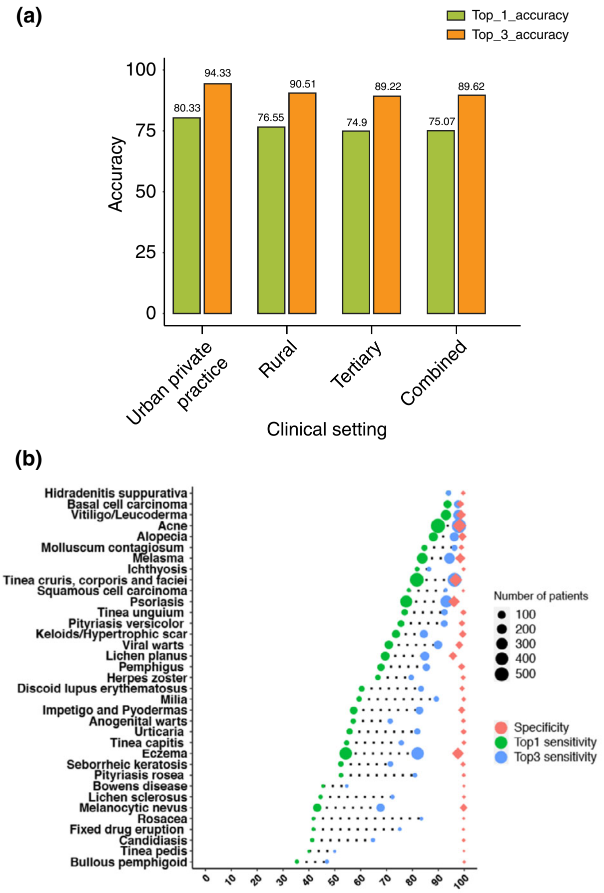
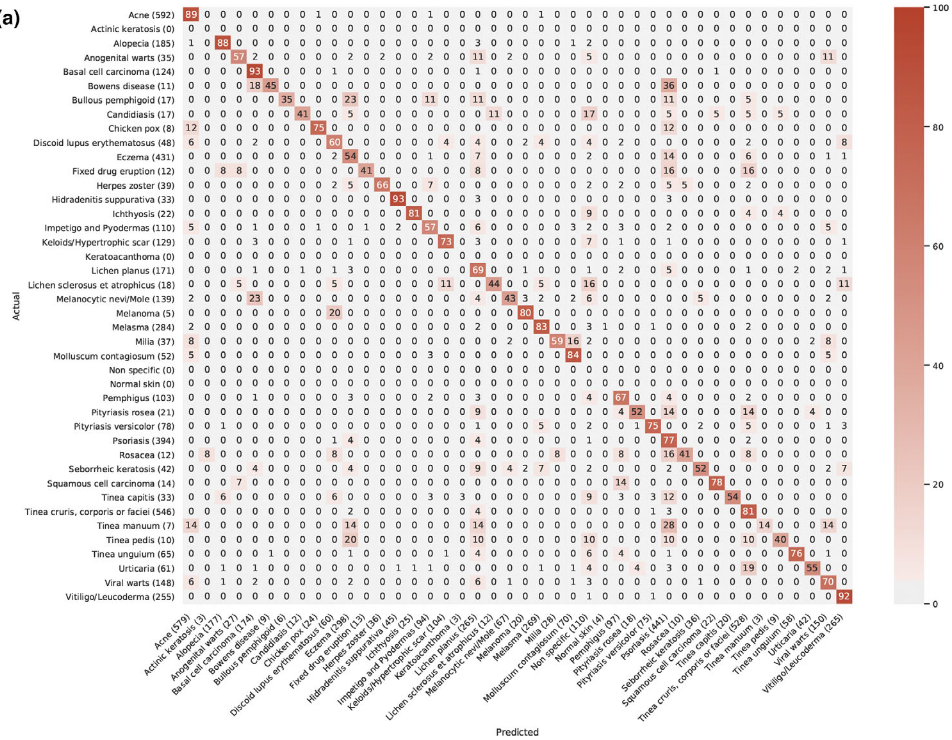


Figure 3 (a) Overall Top-1 and Top-3 accuracies from clinical validation of app in a private urban clinic $N = 383$), rural hospital ($N = 932$), tertiary hospital ($N = 3699$) and combined data ($N = 5014$). (b) Disease-specific Top-1 sensitivities, Top-3 sensitivities and Top-1 specificities. The length of the dotted line depicts an increase in sensitivity between first and third predictions of the app. The size of the icons depicts the number of patients for that disease class.

(a)



(b)

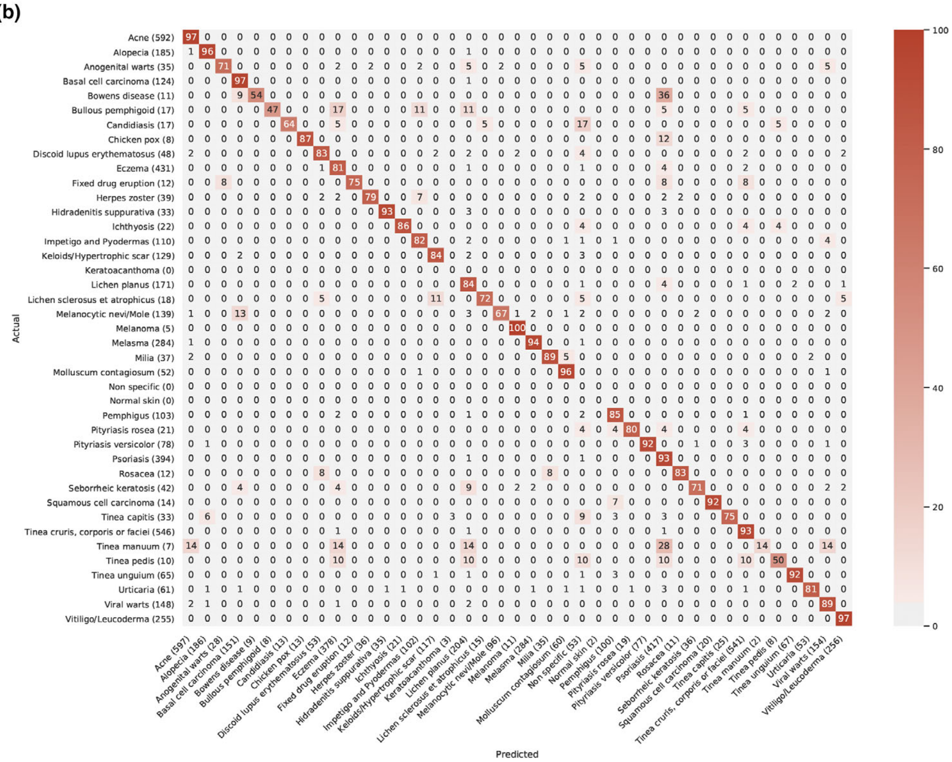


Figure 4 (a) Confusion matrix from the top-1 prediction of clinical data by the app. The numbers in each row are normalized to 100. The actual patient numbers for each disease class are shown in parenthesis. The actual labels are depicted on the y-axis and predicted labels are on the x-axis. (b) Confusion matrix from the top-3 prediction of clinical data by the app. The numbers in each row are normalized to 100. The actual patient numbers for each disease class are shown in parenthesis.

to diagnose 26 common skin conditions from clinical images plus associated medical histories.[25] During validation, Liu *et al* achieved a 71% top-1 accuracy and their algorithm surpassed dermatologists, PCPs and nurse practitioners in a clinical dataset.[25] Our Top-1 accuracy was 77% and the mean AUC was 0.95 for algorithmic validation of 40 diseases.

There are no precedents for studies in which an AI-based, skin disease diagnosis app has been tested in clinical settings. In our work, a top-1 accuracy of 75% and a top-3 accuracy of 90% was achieved by the app for 35 skin diseases and was largely similar to our model validation results. Some differences in diagnostic performance (in terms of disease AUCs) between model and app were noted and may be attributable to either fewer clinical validation images for some diseases like Bowen's disease, anogenital warts, tinea pedis, rosacea, candidiasis, herpes zoster (Fig. 4a and Supplementary Figure S1) and/or differences in disease morphology between white and pigmented skin. Eczema is a broad disease category, incorporating diseases with varied morphology such as lichen simplex chronicus, atopic dermatitis, nummular eczema and contact dermatitis. Training of the model with specific disease classes rather than a broad category of eczema may improve sensitivity. As shown in the confusion matrix, eczema was often confused with psoriasis, lichen planus and tinea corporis, cruris or faciei (Figs 4a, b). To some extent, this mirrors the diagnostic confusion faced by many physicians as well.[2,25,26] A few melanocytic nevi lesions were mistaken for BCC possibly because of their resemblance with pigmented BCC, which is more commonly seen in skin of colour. Besides, it was observed that diseases with distinct morphology and predilection for certain sites (in a similar recognizable background) such as melasma and tinea unguium had better accuracies.

In this study, we found a high top-3 sensitivity for BCC (97.58%, $n = 124$) and SCC (92.86%, $n = 14$) and melanomas (100%, $n = 5$) (Fig. 4b). While the number of skin cancer patients evaluated in this study is relatively small, it is important to note that the skin cancer incidence in India is low. The greatest global skin disease burden is for diseases such as eczema, acne, psoriasis, urticaria, viral diseases and fungal infections,[2] all of which are addressed by the app.

Another interesting application is in the differentiation of tinea corporis and cruris from eczema. There is an alarming epidemic of chronic, treatment-resistant, steroid-modified dermatophyte infection in India.[27] It is mainly attributable to the rampant prescription of corticosteroid creams, by pharmacists and general physicians, which exacerbates this infection.[28,29] In this study, the top-1 and top-3 sensitivity, and specificity of tinea cruris, corporis or faciei diagnosis were 82%, 96% and 97%, respectively and only 11 cases out of 369 tinea cruris, corporis or faciei cases (2% cases) were misdiagnosed as eczema suggesting that this tool could be useful in mitigating this crisis (Fig. 4a).

In this work, clinical images of the patients' lesions were taken by seven different dermatologists in primary, tertiary and urban practice in settings of varying lighting conditions, backgrounds, with mobile phones having different camera settings, resolution, zoom and focus. However, there was not much difference in the accuracy of the app in three different settings suggesting that such an app can be a robust clinician-friendly technology under varying conditions.

The scarcity of training data, especially clinical images from different ethnicities, is considered the biggest challenge in developing optimal deep learning-based skin disease classifiers. The public dermatology archives like ISIC largely contain skin cancer dermoscopic images rather than clinical images of common diseases and are highly skewed in terms of image numbers for different diseases. With the use of locally generated data, we were able to counter the issue of class imbalance to some degree by selective collection from patients. Also, dataset bias and racial bias have been described, arising from the reuse of the same public datasets from white-skinned individuals for training, evaluating or comparing models in nearly all studies.[30] Therefore, we used several thousand new patient images for training and all distinct images for validation. To enable our app to make decisions on clinically relevant features, we also trained our model on images of normal skin from individuals as a control.

One limitation of this study was the absence of integration of medical history with image analysis by the app. The dermatologists had the advantage of having access to relevant additional information as they had a face-to-face interaction with the patient, making it easier for them to arrive at a diagnosis. A few important relevant points in the analysis, such as the duration of the lesion, sites(s) involved, symptoms, and evolution of the lesions can improve the accuracy. The image analysis does not take into consideration the distribution of skin lesions, which is as important a clue for the diagnosis of certain diseases as is the morphology. For example, pityriasis rosea and herpes zoster have a patterned distribution. Future versions of such apps need to refine the image-based diagnosis with automated analyses of patient metadata.

In conclusion, this is the first study where an app designed to detect 40 dermatological diseases was tested in actual clinical settings on patients with skin of colour. Our data suggest that the AI-driven app has high diagnostic accuracy compared to a dermatologist, and is, therefore, a useful, point-of-care, clinical decision support tool for dermatological diagnosis for a range of common skin conditions.

References

1 Hollestein LM, Nijsten T. An insight into the global burden of skin diseases. *J Invest Dermatol* 2014; **134**: 1499–1501.

2 Karimkhani C, Dellavalle RP, Coffeng LE *et al*. Global skin disease morbidity and mortality: an update from the global burden of disease study 2013. *JAMA Dermatol* 2017; **153**: 406–412.

3 Skin diseases to grow in India by 2015: Report, May 2014. (https://www.biospectrumindia.com/news/73/8437/skin-diseases-to-grow-in-india-by-2015-report.html)Accessed on April 10, 2020.

4 Yoo JY, Rigel DS. Trends in dermatology: geographic density of US dermatologists. *Arch Dermatol* 2010; **146**: 779.
5 Ehrlich A, Kostecki J, Olkaba H. Trends in dermatology practices and the implications for the workforce. *J Am Acad Dermatol* 2017; **77**: 746–752.
6 Lim HW, Collins SAB, Resneck JS, Jr *et al*. The burden of skin disease in the United States. *J Am Acad Dermatol* 2017; **76**: 958–972.e2.
7 Patro BK, Tripathy JP, De D, Sinha S, Singh A, Kanwar AJ. Diagnostic agreement between a primary care physician and a teledermatologist for common dermatological conditions in North India. *Indian Dermatol Online J* 2015; **6**: 21–26.
8 Bahelah SO, Bahelah R, Bahelah M, Albatineh AN. Primary care physicians' knowledge and self-perception of competency in dermatology: An evaluation study from Yemen. *Cogent Med* 2015; **2**: 1119948.
9 Ramsay DL, Fox AB. The ability of primary care physicians to recognize the common dermatoses. *Arch Dermatol* 1981; **117**: 620–622.
10 Tran H, Chen K, Lim AC, Jabbour J, Shumack S. Assessing diagnostic skill in dermatology: a comparison between general practitioners and dermatologists. *Australas J Dermatol* 2005; **46**: 230–234.
11 Pickett K, Loveman E, Kalita N, Frampton GK, Jones J. Educational interventions to improve quality of life in people with chronic inflammatory skin diseases: systematic reviews of clinical effectiveness and cost-effectiveness. *Health Technol Assess* 2015; **19**: 1-176,v-vi.
12 Gordon WJ, Landman A, Zhang H, Bates DW. Beyond validation: getting health apps into clinical practice. *NPJ Digit Med* 2020; **3**: 14.
13 Newzoo's Global Mobile Market Report: Insights into the World's 3.2 Billion Smartphone Users, the Devices They use & the Mobile Games They Play. (https://newzoo.com/insights/articles/newzoos-global-mobile-market-report-insights-into-the-worlds-3-2-billion-smartphone-users-the-devices-they-use-the-mobile-games-they-play/)(Accessed on April 10, 2020).
14 Sanders SF, Terwiesch M, Gordon WJ, Stern AD. How artificial intelligence is changing health care delivery. *N Engl J Med Catalyst* 2019. (https://catalyst.nejm.org/health-care-ai-systems-changing-delivery/) (Accessed: April 10, 2020).
15 Chuchu N, Takwoingi Y, Dinnes J *et al*. Cochrane Skin Cancer Diagnostic Test Accuracy Group. Smartphone applications for triaging adults with skin lesions that are suspicious for melanoma. *Cochrane Database Syst Rev* 2018; **12**: CD013192.
16 Huang G, Liu Z, MaatenLvd WKQ.Densely connected convolutional networks 2017 IEEE conference on computer vision and pattern recognition (CVPR), Honolulu, HI, 2017;2261–9. (http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8099726&isnumber=8099483) (Accessed on April 10, 2020).
17 Huang G, Liu Z, van der Maaten L, Weinberger KQ.Densely connected convolutional networks. August 2016(arXiv:1608.06993[cs.CV]).
18 Lin T-Y, Goyal P, Girshick R, He K, Dollár P.Focal Loss for Dense Object Detection. August 2017 (arXiv:1708.02002 [cs.CV]).
19 Log loss. February 2017 (http://wiki.fast.ai/index.php/Log_Loss). Accessed on: May 13, 2020.
20 Pedregosa F, Varoquaux G, Gramfort A *et al*. Scikit-learn: machine learning in Python. *J Mach Learn Res* 2011; **12**: 2825–2830.
21 Topol R. Deep Medicine: how artificial intelligence can make healthcare human again Ch, 5th edn, Basic Books, New York, 2019.
22 Esteva A, Kuprel B, Novoa RA *et al*. Dermatologist-level classification of skin cancer with deep neural networks. *Nature* 2017; **542**: 115–118.
23 Mendes DB, da Silva NC.Skin lesions classification using convolutional neural networks in clinical images, December 2018(arXiv:1812.02316 [cs.CV])(Last accessed: April 10, 2020).
24 Han SS, Kim MS, Lim W, Park GH, Park I, Chang SE. Classification of the clinical images for benign and malignant cutaneous tumors using a deep learning algorithm. *J Invest Dermatol* 2018; **138**: 1529–1538.
25 Liu Y, Jain A, Eng C*et al*.A deep learning system for differential diagnosis of skin diseases, September 2019(arXiv:1909.05382) (Accessed on April 2020).
26 Gorouhi F, Davari P, Fazel N. Cutaneous and mucosal lichen planus: a comprehensive review of clinical subtypes, risk factors, diagnosis, and prognosis. *Scien World J* 2014; **2014**: 742826.
27 Siegfried EC, Hebert AA. Diagnosis of atopic dermatitis: mimics, overlaps, and complications. *J Clin Med* 2015; **4**: 884–917.
28 Al Hasan M, Fitzgerald SM, Saoudian M, Krishnaswamy G. Dermatology for the practicing allergist: Tinea pedis and its complications. *Clin Mol Allergy* 2004; **2**: 5.
29 Verma SB, Vasani R. Male genital dermatophytosis– clinical features and the effects of the misuse of topical steroids and steroid combinations – an alarming problem in India. *Mycoses* 2016; **59**: 606–614.
30 Bissoto A, Fornacialli M, Valle E, Avila S.(De)Constructing bias on skin lesions datasets. April 2019 (arXiv:1904.08818) (Accessed on April 2020).

Supporting information

Additional Supporting Information may be found in the online version of this article:

Supplementary Material

Supplementary Figure 1. ROC curve of Machine and clinical validation studies of individual disease classes.

Supplementary Methods: Formulae of the metrics used in the study.